



AgriRobot

AI-powered agricultural robot for automated nutrient-deficiency detection and targeted fertilizer spraying.

KAHNDSE25.2F-005

KAHNDSE25.2F-014

KAHNDSE25.2F-031

KAHNDSE25.F-032

D.G.C Minidu

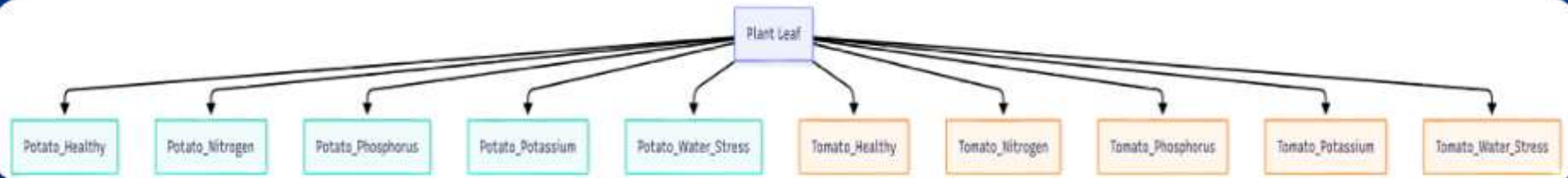
J.P.K.S Lakmal

K.P.G.M.M Rajapaksha

Dineth Weerasekara

Project-Data-Set

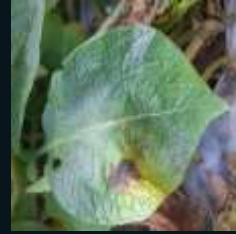
Plant Leaf Data-Set



Clarification

- 1000 images per each Batch Class (10,000 images)
- Previous shortages of images corrected by obtaining real images from Nuwara Eliya
- Machine Training using Google Collab with T4 Gpu

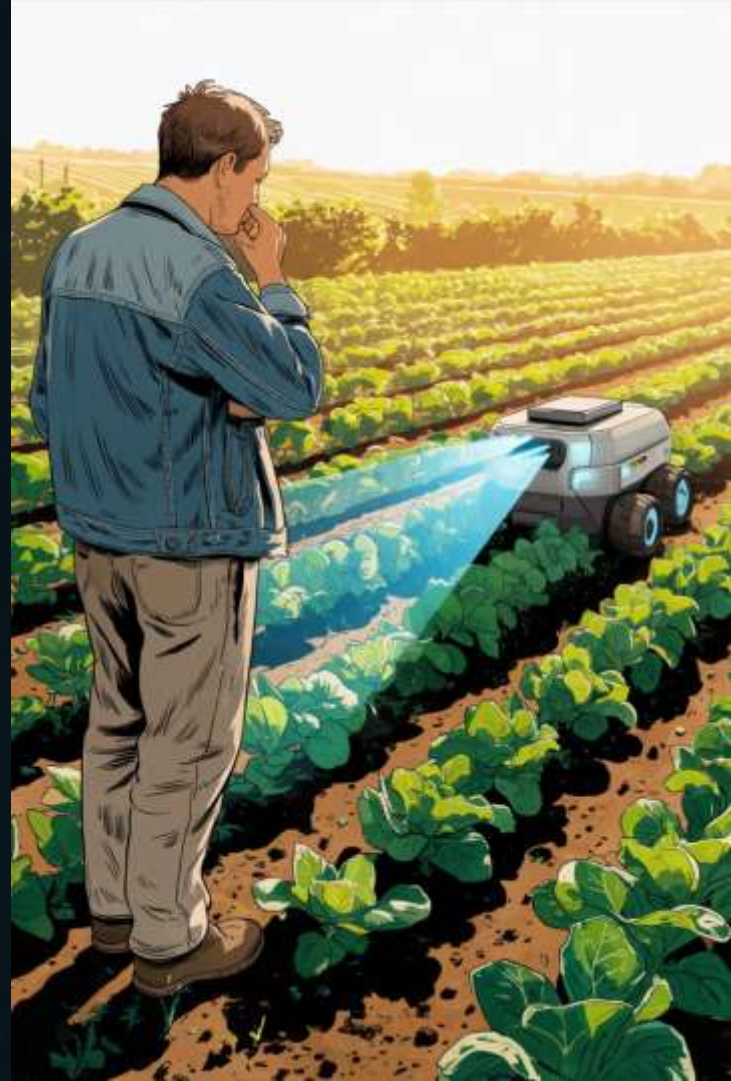
Plant Leaf Data-Set



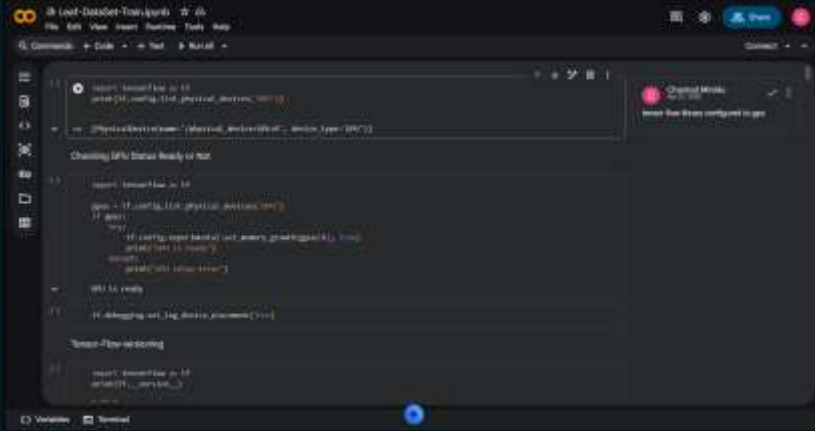
Potato_Nitrogen

Potato_Phosphorus

Tomato_Nitrogen

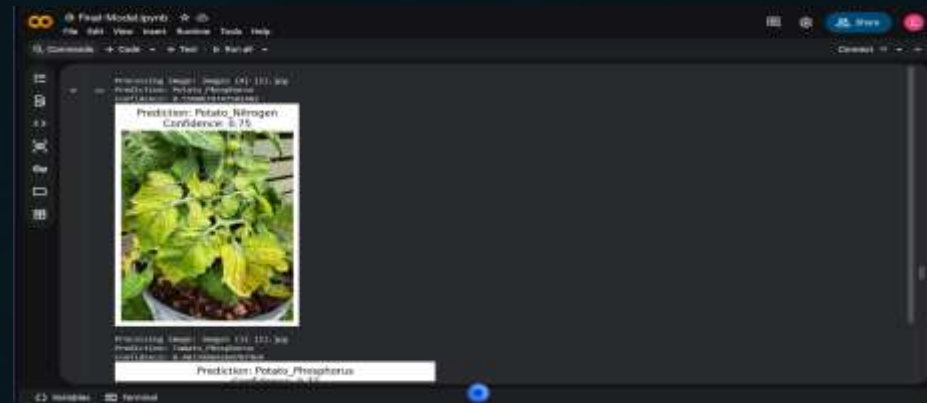


Machine Learning



```
1 import tensorflow as tf
2 print(tf.config.list_physical_devices('GPU'))
3
4 # Physical device(s) name: '/physical_device:/gpu:0', device type: 'GPU'
5
6 Checking GPU Status ready to run
7
8 import tensorflow as tf
9
10 gpus = tf.config.list_physical_devices('GPU')
11
12 try:
13     if len(gpus) > 0:
14         tf.config.set_logical_device_configuration(
15             gpus[0], tf.config.LogicalDeviceConfiguration(accelerator_index=0))
16         print('GPU is ready')
17     else:
18         print('No GPU found')
19
20 # TensorFlow will log device placement info
21
22 Tensorflow version
23
24 import tensorflow as tf
25 print(tf.__version__)
```

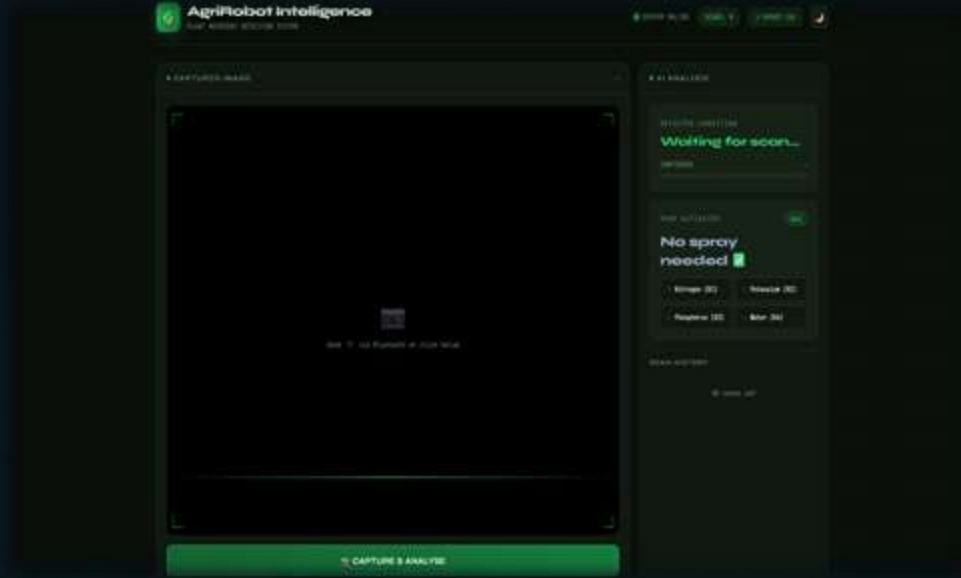
Machine Learning Process :- Google Collab



```
1 # Predicting Image: Image (0) [11, 30]
2 Predictions: Potato_Phenomenon
3 Confidences: 0.75
4
5 Predicting Image: Image (1) [11, 30]
6 Predictions: Potato_Phenomenon
7 Confidences: 0.75
8
9 Predicting Image: Image (2) [11, 30]
10 Predictions: Potato_Phenomenon
11 Confidences: 0.75
```



Esp-32 Cam-Server & Dashboard



Toggle Light/Dark

Toggle button for Light and Dark Mode for the interface



Overview

Indicates the Total Scans, Confidence, Healthy Plants & Deficiencies Found



Confidence Trends

Confidence Scores (Avg), Minimum, Medium, Max



Distribution

Indicates the Distribution on Condition and the Crops

- Model Performance Metrics:-
Model_info
- Per Class accuracy

ML Model — Specs & Actions

- **Server.py** acts as the brain of the system
- **ML Inference** — Loads a TFLite model (`final_model.tflite`) that classifies plant images into 10 categories: Healthy, Nitrogen deficiency, Phosphorus deficiency, Potassium deficiency, and Water Stress — for both Potato and Tomato crops.
- **ESP32-CAM Integration** — Fetches JPEG images from the camera module over the local network via HTTP.
- **Pump Control** — Maps each detected condition to a specific relay (1–4), triggering the correct pump (Nitrogen, Phosphorus, Potassium, or Water) for 15 seconds automatically in a background thread.



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Classes & Pump Mapping

Potato/Tomato × {Healthy, Nitrogen, Phosphorus, Potassium, Water Stress}

specific pumps: Pump1 GPIO16 (N), Pump2 GPIO17 (K), Pump3 GPIO18 (P), Pump4 GPIO19 (Water).

Dashboard Structure



Header

Agri-Robot logo with a live server status indicator
Total scan counter showing how many analyses have been performed



Live Camera Stream Panel

Displays the real-time MJPEG stream from the ESP32-CAM



Capture image Panel

Displays the most recently captured and analyzed plant image



AI-Analysis Panel

Detected Condition card —
Shows the predicted class (e.g. "Tomato Nitrogen") with a color-coded confidence bar that animates on update



Scan History

A scrollable list of the last 10 scans with timestamp, detected condition, confidence percentage, and pump triggered

Data-Base :-My SQL Workbench

Table	What it Stores
Scans	Every AI prediction — crop, condition, confidence, image, pump triggered
Pump_events	Every pump activation — which relay, how long it sprayed
Sensor_readings	DHT-22 temperature & humidity + soil moisture percentage <i>-future</i>
alerts	Repeated deficiency warnings triggered by the dashboard
sessions	Server start/stop tracking

Database is future-ready — sensor tables are pre-built to accept DHT-22 and Capacitive Soil Moisture data without any schema changes.

Data-Base Integration

- ESP32-CAM
- ↓ JPEG image
- Flask Server (server.py)
- ↓ TFLite inference
- AI Prediction + Confidence
- ↓ save_scan()
- MySQL Database ←——
(pump_events, alerts,
 sensor_readings)
- ↓
- Dashboard (live queries)



Fertilizers Used

Blue Crystal Fertilizer

This generally refers to **NPK compound fertilizer**. People call it “Blue Crystal” because of its blue color. It contains the three main nutrients essential for plant growth:

Nitrogen (N) – for leaf growth

Phosphorus (P) – for root and flower development

Potassium (K) – for fruit formation and disease resistance

Orange crystal Fertilizer

Triple Super Phosphate(TSP)

Muriate of Potash (MOP)

Urea

Urea – for nitrogen supply





Conclusion & Next Steps

Key Achievements

Built a Bluetooth-controlled 4-wheel robot, full AI pipeline (capture → TFLite → pump), and a real-time monitoring dashboard with modular design for sensors & crops.

Challenges

Relay power issue (used 3.3V instead of required 5V), ESP32-CAM IP changes, and dashboard refresh on repeated same-class scans — root causes identified and mitigations applied.

Next Steps

Correct relay wiring, install sensors, mount tanks & pumps, conduct full field trials on potato & tomato, calibrate spray durations, and finalize system integration for demonstration.

Thank you — Questions & Discussion